

```
!pip install gradio
```

```
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.50.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.12.1)
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Requirement already satisfied: groovy~0.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)
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```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Static classifier
from sklearn.linear_model import LogisticRegression

# Boosting models
from sklearn.ensemble import AdaBoostClassifier
from sklearn.ensemble import GradientBoostingClassifier

# Ensemble method
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import VotingClassifier

from sklearn.metrics import accuracy_score
import gradio as gr
```

```
df = pd.read_csv("diabetes.csv")

print(df.head())
print(df.shape)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6

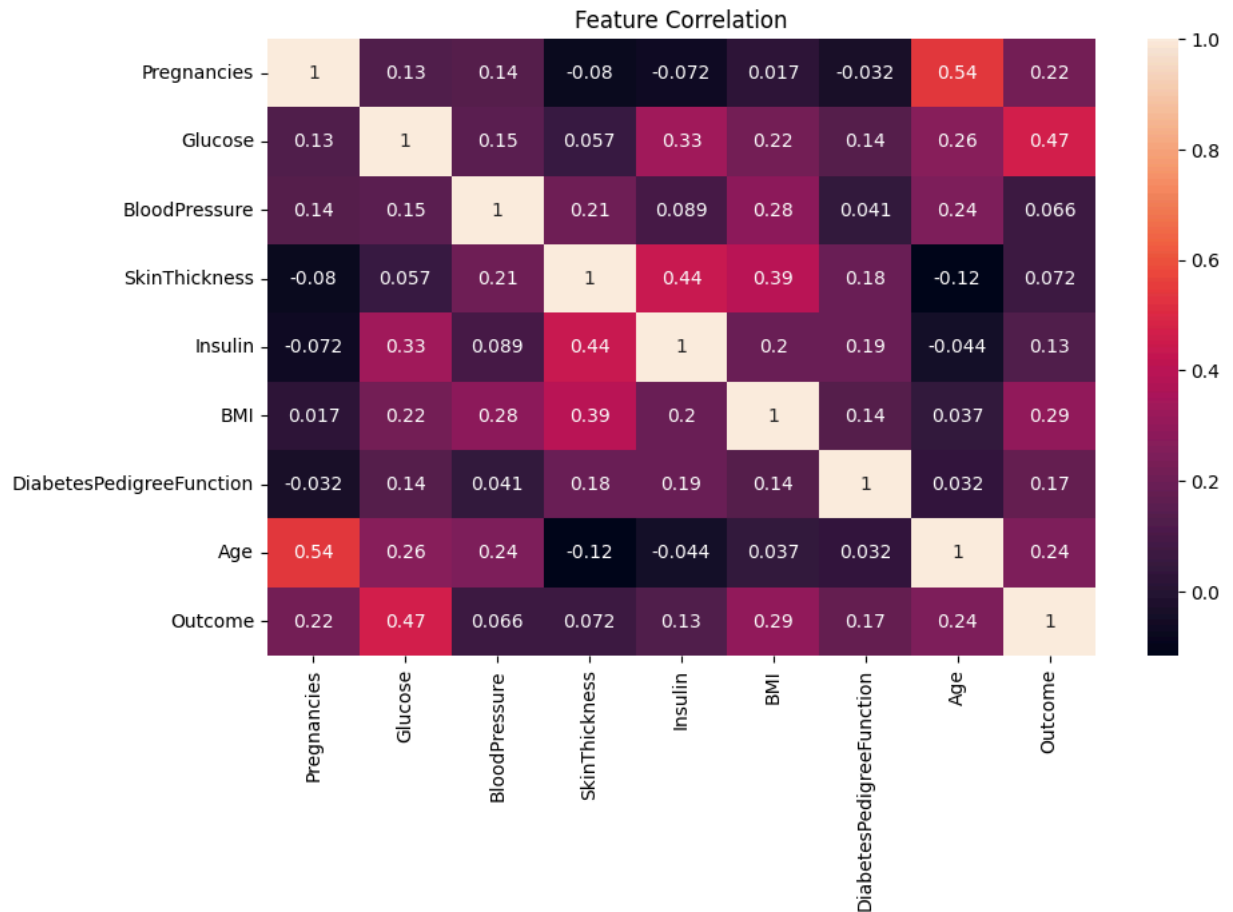
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

(769, 9)

```
import seaborn as sns
```

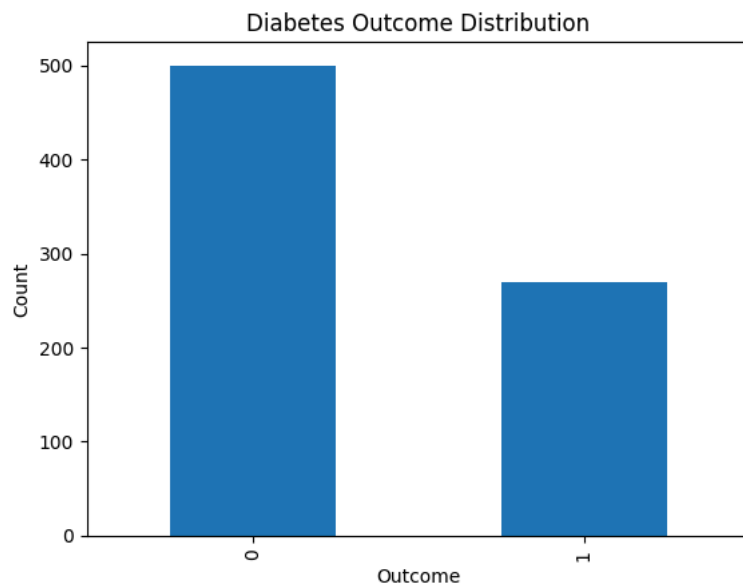
```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True)
plt.title("Feature Correlation")
plt.show()
```



```
df['Outcome'].value_counts().plot(kind='bar')
```

```
plt.title("Diabetes Outcome Distribution")
plt.xlabel("Outcome")
plt.ylabel("Count")

plt.show()
```



```
X = df.drop("Outcome", axis=1)
y = df["Outcome"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
log_model = LogisticRegression()

log_model.fit(X_train, y_train)

pred1 = log_model.predict(X_test)

print("Logistic Regression Accuracy:",
      accuracy_score(y_test, pred1))
```

Logistic Regression Accuracy: 0.7662337662337663

```
ada = AdaBoostClassifier()

ada.fit(X_train, y_train)

pred2 = ada.predict(X_test)

print("AdaBoost Accuracy:",
      accuracy_score(y_test, pred2))
```

AdaBoost Accuracy: 0.7727272727272727

```
gb = GradientBoostingClassifier()

gb.fit(X_train, y_train)

pred3 = gb.predict(X_test)

print("Gradient Boosting Accuracy:",
      accuracy_score(y_test, pred3))
```

Gradient Boosting Accuracy: 0.7662337662337663

```
rf = RandomForestClassifier()

rf.fit(X_train, y_train)

pred4 = rf.predict(X_test)

print("Random Forest Accuracy:",
```

```
accuracy_score(y_test, pred4))
```

Random Forest Accuracy: 0.7597402597402597

```
rf = RandomForestClassifier()

rf.fit(X_train, y_train)

pred4 = rf.predict(X_test)

print("Random Forest Accuracy:",
      accuracy_score(y_test, pred4))
```

Random Forest Accuracy: 0.7662337662337663

```
ensemble = VotingClassifier(
    estimators=[
        ('lr', log_model),
        ('rf', rf),
        ('ada', ada)
    ],
    voting='hard'
)

ensemble.fit(X_train, y_train)

pred5 = ensemble.predict(X_test)

print("Voting Ensemble Accuracy:",
      accuracy_score(y_test, pred5))
```

Voting Ensemble Accuracy: 0.7662337662337663

```
def predict_diabetes(preg, glu, bp, skin, ins, bmi, dpf, age):

    data = np.array([[preg, glu, bp, skin, ins, bmi, dpf, age]])

    data = scaler.transform(data)

    pred = ensemble.predict(data)

    if pred[0] == 1:
        return "High Risk of Diabetes"
    else:
        return "Low Risk of Diabetes"
```

```
interface = gr.Interface(

    fn=predict_diabetes,

    inputs=[
        gr.Number(label="Pregnancies"),
        gr.Number(label="Glucose"),
        gr.Number(label="Blood Pressure"),
        gr.Number(label="Skin Thickness"),
        gr.Number(label="Insulin"),
        gr.Number(label="BMI"),
        gr.Number(label="Diabetes Pedigree Function"),
        gr.Number(label="Age")
    ],

    outputs="text",

    title="Diabetes Prediction System",

    description="Enter patient medical details to predict diabetes risk"
)

interface.launch()
```

It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://760025351ad10c20e5.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the



No interface is running right now